Limits – 2

Recurrences solvable in closed form

Homework

1 сентября 2026 года

Theoretical minimum. Linear recurrences and the characteristic polynomial; multiple roots and an inhomogeneous right-hand side; generating functions and partial fractions; conjugate surds $\alpha^n + \beta^n$; the matrix form $\binom{a_{n+1}}{a_n} = A\binom{a_n}{a_{n-1}}$; proofs of integrality by induction and by substitution.

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1. Let $x_n = \frac{(x_{n-1})^2-9}{8}$ for $n \geqslant 2$. Find $\lim_{n\to\infty} x_n$ for all initial values $x_1 \in \mathbb{R}$.

2. Let t and ε be real with $|\varepsilon| < 1$. Prove that the equation $x - \varepsilon \sin x = t$ has a unique real solution, and write it as the limit of an explicit iteration.

3. (IMC 2015) Let $F(0) = 0$, $F(1) = \frac{3}{2}$ and $F(n) = \frac{5}{2}F(n-1) - F(n-2)$ for $n \geqslant 2$. Determine whether $\sum_{n\geqslant 0} \frac{1}{F(2^n)}$ is rational.

4. (IMC 2019) Define $a_0 = 1$, $a_1 = 2$ and $(n+3)a_{n+2} = (6n+9)a_{n+1} - na_n$ for $n \geqslant 0$. Prove that all terms of the sequence are integers.

5. (IMC 2003) Let $a_0 = 1$ and $a_{n+1} = \frac{1}{n+1}\sum_{k=0}^{n} \frac{a_k}{n-k+2}$. Find $\lim_{n\to\infty} \sum_{k=0}^{n} \frac{a_k}{2^k}$.

6. (IMC 2010) Compute the sum of the series $\sum_{k=0}^{\infty} \frac{1}{(4k+1)(4k+2)(4k+3)(4k+4)}$.

7. (IMC 2010) Define $x_1 = \sqrt{5}$ and $x_{n+1} = x_n^2 - 2$ for $n \geqslant 1$. Compute $\lim_{n\to\infty} \frac{x_1 x_2 \cdots x_n}{x_{n+1}}$.

8. (IMC 2008) Let n be a positive integer. Prove that 2^{n-1} divides $\sum_{0\leqslant k<n/2} \binom{n}{2k+1} 5^k$.

9. (IMC 2022) Let p be a prime. A flea sits at the point 0 of the real line. Each minute it either stays where it is, or jumps by 1 to the left or to the right. Let $f(p)$ be the number of ways for the flea to be back at 0 after $p-1$ minutes. Find $f(p)$ modulo p.

10. (AKHIMO 2025) Define $a_1 = 2$, $a_2 = 5$ and $a_{n+2} = f(a_{n+1}, a_n)$ for $n \geqslant 1$, where $f(x, y) = 5(x + y) + 2\sqrt{6x^2 + 15xy + 6y^2}$. Show that a_n is an integer for every n.

11. (OMOUS 2026) Let $a_0 = 1$, $a_1 = 2$ and $a_{n+1} = \frac{3^{n-1} + a_n^2}{a_{n-1}}$ for $n \geqslant 1$.

    a) Show that all a_n are integers.

    b) Does the series $\sum_{n \geqslant 0} \frac{1}{a_n}$ converge?

    c) Is the sum $\sum_{n \geqslant 0} \frac{2^n}{a_{2^n}}$ rational?